



## DSA Master

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## Interview Master

Welcome, Interview Masters!

Today we have curated a list of the *Top Data Structures and Algorithms* to study for interviews.

These concepts will teach you the core techniques you need to know to crush any technical interview.

Algorithms are divided into 2 broad categories:

1. Data Structure specific algorithms
2. General algorithms/techniques

Implement all the algorithms unless otherwise specified.

We will be using some of these algorithms in next editions of Interview Master (if not already covered in the [previous editions](#)).

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### THE ULTIMATE DSA GUIDE

## Data Structure specific algorithms

### 1. Arrays

- Sorting:
  - QuickSort: Efficient average-case time complexity ( $O(n \log n)$ )
  - MergeSort: Stable sort, useful when order matters ( $O(n \log n)$ )
- Searching:
  - Binary Search: Fast search in sorted arrays ( $O(\log n)$ )
- Two Pointers:
  - In-place manipulation, often for sorted arrays (e.g., removing duplicates)
- Sliding Window:
  - Subarray problems, finding maximum/minimum within a window

### 2. Linked Lists

- Traversal:
  - Iterate through the list, understand the node structure
- Insertion/Deletion:
  - At beginning, end, or at a specific position
- Reversal:
  - In-place reversal, recursive and iterative approaches
- Cycle Detection:
  - Floyd's Tortoise and Hare algorithm

### 3. Hash Tables (Hash Maps/Sets)

- Implementation not needed. Just understand following:
  - Understand how hash functions work
  - Insertion/Deletion/Lookup
  - Collision Handling

### 4. Trees (Binary Trees, Binary Search Trees, etc.)

- Traversal:
  - Inorder, Preorder, Postorder (recursive and iterative)
- Searching:
  - Find a node with a given value (especially in BSTs)

### 5. Stacks

- Implementation not needed. Just understand following:
  - Push/Pop/Peek Operations

## 6. Queues

- Implementation not needed. Just understand following:
  - Enqueue/Dequeue Operations

## 7. Heaps (Priority Queues)

- Implementation not needed. Just understand following:
  - Insertion/Deletion (extract-min/max)
  - Building a Heap
- Top K Elements:
  - Using a heap to find k largest/smallest elements

## 8. Graphs

- Traversal:
  - Breadth-First Search (BFS)
  - Depth-First Search (DFS)
- Shortest Path:
  - Dijkstra's Algorithm
- Cycle Detection:
  - DFS

## 9. Tries

- Implement Trie from scratch
- Insertion/Searching:
  - For words/prefixes
- Autocompletion:
  - Using a trie for word suggestions

## 10. Union-Find (Disjoint Set)

- Implement Union-Find from scratch
- Find/Union Operations
- Cycle Detection in undirected graphs

## General algorithms/techniques

### 1. Recursion

- Defining a problem in terms of itself, often leading to elegant and concise solutions.
- Solve: Factorial calculation, tree traversals, depth-first search.

### 2. Dynamic Programming

- Breaking down a problem into overlapping subproblems and storing solutions to avoid recomputation.
- Solve: Fibonacci sequence, Knapsack problem, Longest Common Subsequence.

### 3. Greedy Algorithms

- Making locally optimal choices at each step with the hope of finding a global optimum.
- Implement: Kruskal's algorithm for minimum spanning trees.

### 4. Backtracking

- Incrementally building solutions, exploring all possible paths, and abandoning invalid ones.
- Solve: Sudoku solver, N-Queens problem, generating permutations.

### WHAT'S NEXT?

Once you have implemented the above algorithms, solve [Interview Master 100](#), which contains top 100 interview problems.

Each problem builds upon previous problems so that you can gradually expand your knowledge as you progress.

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Cheers,

[Sahil](#) + [Sarrah](#)

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